

NEUROPHARMACOLOGICAL ACTIVITY OF AN AYURVEDIC MEDICINE

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ABSTRACT

Nardostachys jatamansi is used in large quantities in preparation of Ayurvedic medicine. The decoction of roots is used for epilepsy, hysteria and weakness. Available report suggest its effect on brain biogenic amines and on seizure threshold. Present study is designed to evaluate its neuropharmacological profile. Gross behavioural study revealed its responsiveness in few neuropharmacological tests.

INTRODUCTION

The potent healing qualities of herbs have been used in different therapeutic philosophies throughout history. Plants are used in Indian Ayurvedic system, in Chinese medicine alongside acupuncture and other techniques. Medicinal herbs are used as a source drugs.

Modification of mood, behaviour and emotion has always been favorite indulgence of mankind. The use of psychoactive drugs evolve along two paths. The first was in the use of drugs of modify normal behaviour and to produce altered states of feelings for religious, ceremonial or recreational purposes. The second was to alleviate mental ailment. Herbal psychotropic drugs do function through biochemical interactions and augment vital processes of body. On the biochemical level, the numerous ingredients involved in a herb work in synergistic way, with elements involved in the process that chemotherapy would not even consider as being active.

Nardostachys jatamansi belongs to Valerianacea is used in large quantities in preparation of Ayurvedic and Unani drugs (Satyavadi *et al.*, 1976). The decoction of roots had been for centuries in Ayurveda for epilepsy, hysteria, syncope and mental weakness (Chopra *et al.*, 1956; Gupta *et al.*, 1966). Ethanolic extracts have been reported as bitter, tonic, stimulant, antiseptic and to abolish tonic extensor response. Present study is planned to ascertain neuropharmacological role of *N. jatamansi*.

MATERIALS AND METHODS

Wistar rats and albino mice of +ve NMRI strain of either sex, weighing (200-400 g and 18-22 g respectively) were used during these experiments. Animals were purchased from Aga Khan Medical University and PCSIR Laboratories Complex, Karachi. Frogs (*Rana tigrina*) used for isolated tissue experiments, were provided by a local supplier.

Mice and rats were used for neuropharmacological screening after acclimatization period of at least five days to the laboratory environment with 12 h light/12 h dark schedule. The animals were housed in standard metal cages and provided food and water *ad-libitum*.

Extract of roots of *N. jatamansi* was provided by Dr. Mansoor Ahmed, Associate Professor, Department of Pharmacognosy, Faculty of Pharmacy, University of Karachi. Extract was given in dose of 200 mg/kg. Locomotor activity evaluated using an activity meter (Columbus International Corporation). Potentiation of pentobarbitone sleeping time was noted by a method of Dandiya and Coilumbine (1959). For antiaggressive activity test was carried according to Yen *et al.*, (1970).

Analgesic activity was noted by method of Ben-Bessat *et al.*, (1959), Sojae-Echaque and Lin (1962) and Turner (1965). Exploratory activity was also recorded (Turner, 1965; Rushton, 1967; Dorr *et al.*, 1970).

For anticonvulsant activity method described by Bianchi and Franceschini (1954) was used. Hypogenic activity was determined by method described by Janssen (1959).

RESULTS

Extract of *N. jatamansi* showed positive responses in few neuropharmacological tests. Non-significant effects were noted in traction test (Mylorelaxant activity), tail flick method and exploratory activity. Extract did not protect death from pentylenetetrazole-induced convulsions and did not prolong pentobarbitone-induced hypnosis. Result of tail flick test and isolation-induced aggression also did not show highly significant effects.

Decreased motor activity, loss of righting reflex, ptosis, passivity and reduction in amphetamine-induced motility was noted after administration of *N. jatamansi*. Positive result was noted in writhing test.

DISCUSSION

N. jatamansi reduced spontaneous motor activity and also decreased amphetamine-induced hypermotility which may be due to increased release of dopamine at nerve endings in the accumbens nucleus (Sharp *et al.*, 1986). Activation of dopamine receptors in the brain, induce stereotyped behaviour

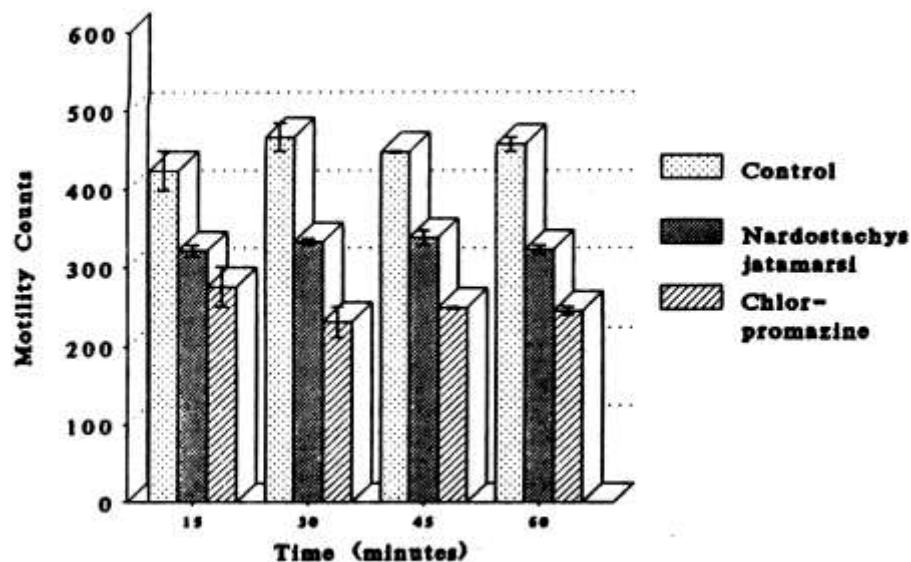


Fig. 1. Effect of *N. jatamansi* on spontaneous motor activity. Each bar represents mean $\pm$ S.E.M. significance of difference was calculated by Mann-whitney U-Test ($P^* < 0.05$)

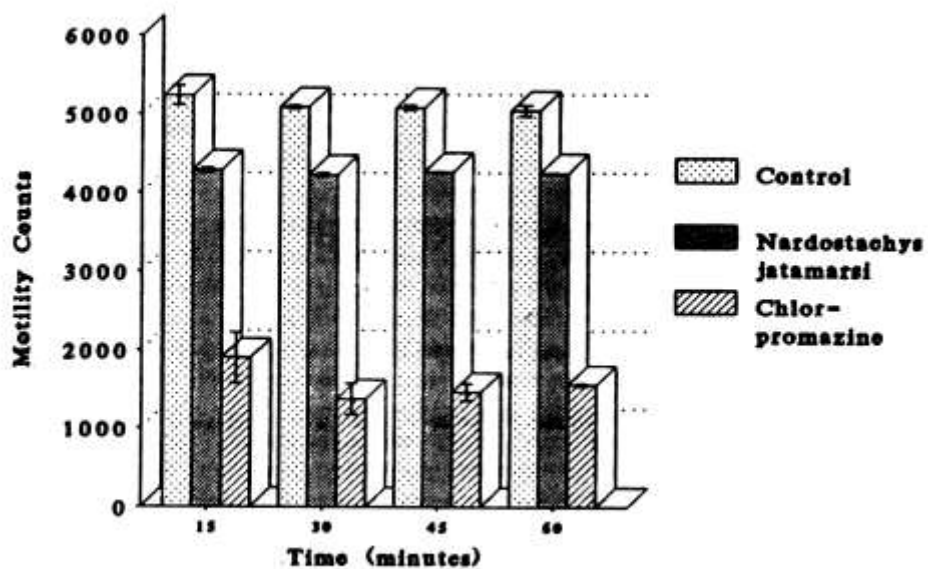


Fig. 2. Effect of crude drug on amphetamine-induced hypermotility. Each bar represents the mean ($n=5$) Significance of difference was calculated by Mann-Whitney U-test ($*P < 0.05$).

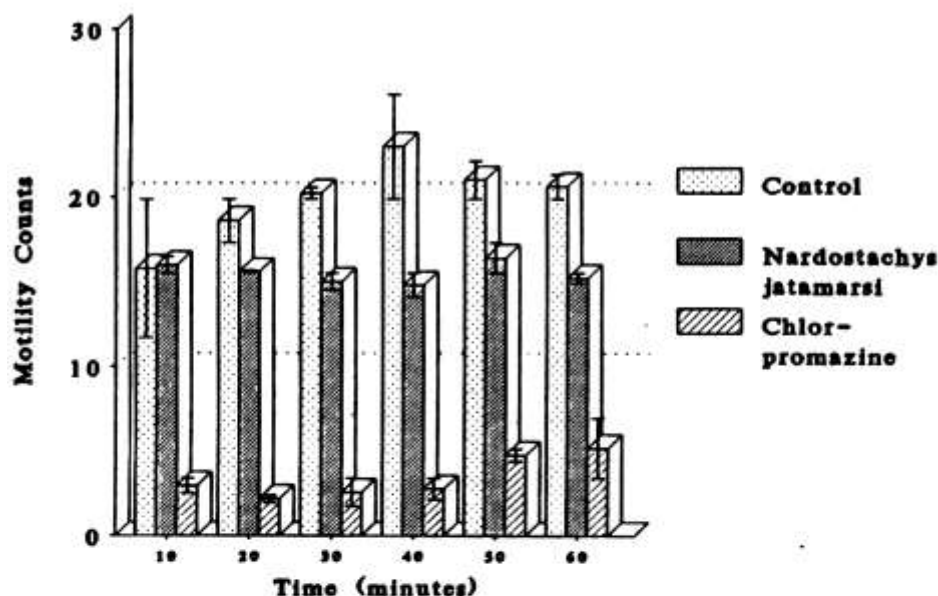


Fig. 3. Values represents the mean $\pm$ S.E.M. ($n=5$). Difference with respect to control calculated by Mann-Whitney U-test (* $P < 0.05$).

and antagonism to these effect indicate neuroleptic activity (Otano *et al.*, 1991).

N. jatamansi is used in large quantities in preparation of Ayurvedic and Unani drugs (Satyavadi *et al.*, 1976). In present report *N. jatamansi* produced significant analgesic effect in writhing test. It decreased spontaneous motor activity, amphetamine-induced hypermotility, rota rod test, overnight isolation-induced aggressiveness. *N. jatamansi* increased seizure threshold against MES (Vidya *et al.*, 1994). Many investigations had been performed to determine the role in the seizure mechanism. Several reports are available on the role of monoamine metabolism (Maynert *et al.*, 1975; Przegalinski, 1985) in experimental as well as clinical seizures. But very few reports are available on the scientific evaluation of crude drugs and natural products on biogenic amines.

Available reports suggest that an increase in the turnover or noradrenaline, DA and 5HT decrease the severity of maximal electroconvulsive seizures (Vidya *et al.*, 1994). The increased seizure threshold by *N. jatamansi* may be due to increased level of biogenic amines i.e., 5HIAA and GABA and taurine in brain. Acute and subchronical administration of *N. jatamansi* extract brought about changes in monoamine as well as inhibitory amino acids. Increased seizure threshold by *N. jatamansi* suggested that a balanced

concentration between different neurotransmitters is essential for increasing the threshold against maximal electroshock seizures. The exact mechanism by which anticonvulsant drugs diminish the frequency of seizures remains unknown. In present report highly significant effects of *N. jatamansi* in different neuropharmacological tests were not evidenced.

REFERENCES

- Ben-Bessat, J., Peretz, e. and Sulman, F.G., (1959). *Arch. Int. Pharmacodyn.*, 122: 343-436. In: A neuropsychopharmacological study of *Syzgium cuminii* (Chakraborty, D., Mahapatra, P.K. and Nagchaudhuri, A.K., eds.) (1986). *Planta Medica*, 2 : 139-143.
- Chopra, R.N., Nayar, S.I. and Chopra, I.C., (1956). *Glossary of Indian Medicinal Plants*, CSIR Publications, New Delhi, p. 44.
- Dandiya, P.C. and Collumbine, H., (1959). *J. Pharmacol. Exp. Ther.*, 125 : 353-359.
- Dorr, M., Joyce, D., Porsolt, R.D., Steinberg, H., Summerfield, A. and Toomkiewicz, M., (1971). *Nature*, 231 : 121-123. In: A neuropsychopharmacological study of *Syzgium cuminii* (Chakraborty, D., Mahapatra, P.K. and Nagchaudhuri, A.K., eds.) (1986). *Planta Medica*, 2 : 139-143.
- Gupta, S.S., Seth, C.B., Mathur, V.S., Ghooi, C. and Verma, R.K., (1966). *J. Physiol. Pharmacol.*, 10 : 5-14.
- Janssen, P.A.J., Jageneau, A. and Niemegeers, C.J.E., (1960). Effects of various drugs on isolation-induced tightening behaviour of male mice, *J. Pharmacol. Exp. Ther.*, 129 : 471.
- Maynert, E.W., Marczynski, T.J. and Browning, R.A., (1975). In: *Advances in Neurology* (Friendlander, W.J., ed.), Raven Press, New York, pp. 79-147.
- Przegalinski, E., (1985). In: *Handbook of Experimental Pharmacology*, Vol. 74, (Frey, H.H. and Janz, D., eds.). Springer-Verlag, Berlin, pp. 101-137.
- Rushton, R., Steinberg, H. and Tinson, C., (1961). *Nature*, 192 : 533-535.
- Satyavadi, G.V. Raina, M.K. and Sharma, M., (1976). In: *Medicinal Plants of India*, Vol. 1, ICMR Publications, New Delhi.
- Turner, R.A., (1965). In: *Screening Methods in Pharmacology*, Academic Press, New York and London, pp. 90, 113-117.
- Vidya, P.M. Karanth, K.S. and Rao, A., (1994). *Planta Medica*, 60 : 114-117.